CLOUD COMPUTING LABORATORY

ASSESSMENT TASK – CASE STUDY

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CASE STUDY 1

Smart Retail Inventory Management System

1. Introduction

A retail chain implements a cloud-based inventory management system to monitor products, sales transactions, stock levels, and demand. Store devices such as Point of Sale (POS) systems and management dashboards communicate with centralized cloud services through APIs.

2. Cloud Architecture

The system consists of the following components:

- **Client Layer:** POS systems and dashboards used by employees and managers.
- **API Layer:** APIs allow clients to communicate with cloud services.
- **Cloud Application Services:** Process inventory and transaction data.
- **Cloud Storage:** Stores product information, transaction logs, and analytics datasets.
- **Platform Services:** Used for data analytics and demand forecasting.
- **Infrastructure:** Provides computing and networking resources.
- **Security Layer:** VPNs, firewalls, authentication, and API gateways protect the system.

3. Working

1. The POS system records a product sale.
2. The transaction is sent to the cloud through an API.
3. Cloud application services process the transaction.
4. Product and transaction information is stored in cloud storage.
5. The inventory level is automatically updated.
6. Analytics services analyze sales data.

7. Demand forecasting predicts future product requirements.
8. Managers can view inventory information through dashboards.
9. API gateways, VPNs, and firewalls secure communication and access.

4. Cloud Services Used

Component	Cloud Technology
POS/Dashboard	Web and client applications
Communication	REST/API services
Storage	Cloud storage
Analytics	Cloud platform services
Computing	Cloud infrastructure
Security	VPN, Firewall, API Gateway

5. Benefits

- Real-time inventory monitoring
- Automatic stock updates
- Scalable cloud infrastructure
- Improved demand forecasting
- Centralized data management
- Secure communication
- Reduced infrastructure maintenance

6. Conclusion

The Smart Retail Inventory Management System demonstrates how infrastructure, platform services, APIs, cloud storage, and web applications can be integrated to provide an efficient and secure cloud-based retail solution.

CASE STUDY 2

Cloud-Based Document Collaboration System

1. Introduction

An enterprise deploys a cloud-based document collaboration system similar to Google Docs. Users can access documents through web browsers and collaborate in real time. Documents are stored in distributed cloud storage with version control.

2. Cloud Architecture

The major components are:

- **Client Layer:** Web browsers and other client devices.
- **Web Application:** Provides document editing and collaboration features.
- **Web APIs:** Synchronize document changes between clients and cloud services.
- **Cloud Storage:** Stores documents and related information.
- **Version Control:** Maintains previous versions of documents.
- **Network Infrastructure:** Provides high availability and low-latency communication.
- **Security:** Uses encryption, identity management, and access permissions.

3. Working

1. A user logs into the cloud document application using a browser.
2. The user opens or creates a document.
3. The document is stored in distributed cloud storage.
4. Multiple users can access the same document.
5. Changes made by one user are transmitted through Web APIs.
6. The changes are synchronized with other connected users.
7. Version control maintains previous versions of the document.
8. Access permissions determine who can view or edit the document.
9. Encryption protects documents during storage and transmission.

4. Cloud Services Used

Component	Cloud Technology
Client	Web Browser
Application	Cloud-based Web Application

Communication	Web APIs
Storage	Distributed Cloud Storage
Synchronization	API-based synchronization
Version Management	Version Control
Security	Encryption and Identity Management

5. Benefits

- Real-time collaboration
- Access from multiple devices
- Automatic document synchronization
- Version history
- High availability
- Centralized document storage
- Secure sharing and access control

6. Conclusion

The Cloud-Based Document Collaboration System demonstrates how cloud storage, web applications, APIs, distributed infrastructure, and security mechanisms can provide real-time and reliable document collaboration.

CASE STUDY 3

Online Food Ordering Platform

1. Introduction

A startup develops a cloud-based online food ordering platform that can be accessed through web browsers and mobile applications. Customers can view menus, place orders, make payments, and receive order information through the cloud platform.

2. Cloud Architecture

The system contains the following components:

- **Client Layer:** Web browsers and mobile applications.
- **Frontend:** Provides the user interface for customers and restaurants.
- **RESTful Web APIs:** Connect frontend clients with backend services.

- **Backend Services:** Process customer orders and other operations.
- **Cloud Storage:** Stores menu images, invoices, and customer information.
- **Virtual Machines:** Provide computing resources.
- **Managed Kubernetes:** Manages containerized services and supports scalability.
- **Security Layer:** Uses HTTPS, authentication tokens, and role-based access control.

3. Working

1. A customer opens the food ordering application.
2. The frontend requests menu information through RESTful APIs.
3. Backend cloud services process the request.
4. Menu information and images are retrieved from cloud storage.
5. The customer selects food items and places an order.
6. The order is sent to backend services through REST APIs.
7. Customer and invoice information is securely stored in the cloud.
8. Virtual machines and Kubernetes provide scalable computing resources.
9. Authentication tokens verify users.
10. HTTPS protects communication between clients and servers.
11. Role-based access control restricts access according to user roles.

4. Cloud Services Used

Component	**Cloud Technology**
Client	Web Browser / Mobile App
Frontend	Cloud Web Application
Communication	RESTful Web APIs
Backend	Cloud Services
Storage	Cloud Storage
Computing	Virtual Machines
Container Management	Managed Kubernetes
Security	HTTPS, Authentication Tokens, RBAC

5. Benefits

- Access through web and mobile devices
- Scalable infrastructure
- Reliable order processing
- Secure customer data storage
- Real-time communication
- High availability
- Easy management of multiple services
- Flexible resource allocation

6. Conclusion

The Online Food Ordering Platform demonstrates the integration of cloud clients, RESTful APIs, cloud storage, virtual machines, managed Kubernetes, and security services to build a scalable and reliable real-time cloud application.

OVERALL CONCLUSION

The three case studies demonstrate different practical applications of cloud computing. The Smart Retail Inventory Management System uses cloud storage, analytics, APIs, and security for inventory management. The Document Collaboration System uses distributed storage, APIs, synchronization, and version control for real-time collaboration. The Online Food Ordering Platform combines APIs, cloud storage, virtual machines, Kubernetes, and security mechanisms for scalable online services.

These case studies show how cloud computing provides **scalability, availability, centralized storage, real-time communication, security, and efficient resource management** for modern applications.